

# Hardware Efficiency & Worker Density

Hosting 100+ Concurrent Digital Employees on a Single Modest Server

PAGE 39 / 42

## Maximizing Hardware Density: The Economics of Efficiency

The true test of enterprise scalability is operational efficiency per server dollar. Because Fathom compiles to lean machine code with zero runtime interpreter overhead, it achieves unprecedented hardware worker density.

### PYTHON FRAMEWORK FOOTPRINT (100 AGENTS)

- **RAM Required:** 64 GB – 128 GB RAM (Heavy Kubernetes Cluster).
- **CPU Saturation:** High context-switching drag and Python GIL contention.
- **Monthly Server Cost:** \$400 – \$1,200 / month on AWS/GCP.
- **Hosting Cost Per Worker:** \$4.00 – \$12.00 / month.

### FATHOM RUST FOOTPRINT (100 COWORKERS)

- **RAM Required:** 1.5 GB – 3.5 GB RAM (Modest 8-Core Box).
- **CPU Utilization:** Zero idle CPU burn; epoll/kqueue event-driven I/O.
- **Monthly Server Cost:** \$30 – \$60 / month on Hetzner/DigitalOcean.
- **Hosting Cost Per Worker:** < \$0.50 / month.

### Server Resource Allocation Model (8 Cores, 32GB RAM)

**100+**  
COWORKERS

**~25 MB**  
RAM / WORKER

**< 1%**  
IDLE CPU

**99.4%**  
COMPUTE MARGIN

**The Bottom Line:** Fathom cuts server infrastructure costs by 95%, allowing businesses to scale their virtual workforce profitably.